MID EXAM

An **instruction** is a low-level command that the processor executes. It tells the ARM processor what specific action to perform, such as arithmetic operations, data movement, branching, and more.

MOV : It can optionally update the condition flags based on the value.

ADD, ADDS, SUB, SUBS: It can optionally update the condition flags based on the result.

MUL: These 32 bits do not depend on whether the source register values are considered to be signed values or unsigned values.

CMP: It updates the condition flags based on the result, and discards the result.

Branch: In ARM Assembly, a branch is an instruction used to change the flow of execution in a program — similar to a jump or goto in high-level languages.

Branch	Condition (Flags)	Meaning
BGT	`Z == 0` and `N == 0 or V`	Branch if **greater than** (signed)
BLT	`N ≠ V`	Branch if **less than** (signed)
BGE	`N == V`	Branch if **greater than or equal** (signed)
BLE	`Z == 1` or `N ≠ V`	Branch if **less than or equal** (signed)
BEQ	`Z == 1`	Branch if **equal**
BNE	`Z == 0`	Branch if **not equal**
BMI	`N == 1`	Branch if **minus** (negative result)

For making a function call in assembly language. Function call uses r0 to r4 GP registers

LDR and STR : These instructions allow ARM assembly programs to interact with data in memory, which is crucial for tasks such as **processing arrays, accessing global variables, and manipulating data** during program execution.

Data Segment:

The data segment in assembly programming is a section of memory where global variables, constants, and static data are stored.

Key Features of the Data Segment:

• Global Variables: Variables declared in the data segment are typically available throughout the entire program.

• Static Data: Data defined in this segment remains in memory throughout the program's execution.

• Constants: You can define read-only constants in the data segment.

• Uninitialized Data: Variables that are declared but not initialized are stored in a separate section, typically the BSS (Block Started by Symbol) segment.

Segments in ARM Assembly:

1. Text Segment (.text): Contains the executable instructions (code) of the program.

2. Data Segment (.data): Contains initialized data (like global variables and constants).

3. BSS Segment (.bss): Contains uninitialized variables or variables initialized to zero.

Common Directives Used in Data Segment:

- .word: Allocates a word (typically 4 bytes on 32-bit systems) for storing integers or addresses.
- .asciz: Allocates a null-terminated string.
- .byte: Allocates a single byte of data.
- .hword: Allocates a halfword (2 bytes).
- .space: Allocates a block of memory with uninitialized data

Function: To use the concept of function in arm assembly, we basically use the **branch**. Specifically, we use BL (Branch with link) opcode when using function.

- org 1000 @program start at memory address 1000/

it is the same as a SUBS instruction, except that the processor discards the result.